

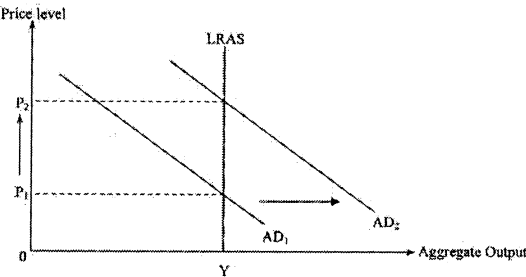
6.2 LONG RUN AND SHORT-RUN EQUILIBRIUM

MARKING SCHEME

1994/AL/II/9 A	2012/DSE/I/34 C (51%)	2014/DSE/I/29 D (69%)	2015/DSE/II/39 B (84%)	2018/DSE/I/26 A (80%)
1997/AL/II/21 D	2012/DSE/I/36 D (52%)	2014/DSE/I/34 C (42%)	2016/DSE/II/26 D (65%)	2018/DSE/I/27 D (76%)
1999/AL/II/4 C	2013/DSE/I/23 D (76%)	2014/DSE/I/35 B (67%)	2016/DSE/II/27 C (57%)	2018/DSE/II/28 B (80%)
2000/AL/II/20 D	2013/DSE/I/24 B (65%)	2014/DSE/I/38 D (75%)	2016/DSE/II/28 B (64%)	2018/DSE/I/38 D (58%)
2001/AL/II/14 A	2013/DSE/I/29 A (56%)	2015/DSE/I/26 A (62%)	2017/DSE/II/28 C (76%)	2018/DSE/I/39 A (65%)
2012/AL/II/30 C (79%)	2013/DSE/I/30 D (59%)	2015/DSE/I/27 D (68%)	2017/DSE/II/29 D (60%)	2020/DSE/I/ C
2012/DSE/II/25 A (72%)	2014/DSE/I/27 A (61%)	2015/DSE/I/37 C (59%)	2017/DSE/I/37 B (66%)	2021/DSE/II/27 D
2012/DSE/I/26 A (73%)	2014/DSE/I/28 C (48%)	2015/DSE/I/38 A (73%)	2017/DSE/II/40 B (45%)	2021/DSE/I/28 C
2022/DSE/I/29 B	2022/DSE/I/36 C	2022/DSE/I/38 A	2024/DSE/II/29 B	2024/DSE/II/30 C
2025/DSE/II/28 D	2025/DSE/I/29 A	2025/DSE/I/30 A		

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

- 2012/DSE/II/10(c)
- People’s consumption expenditure increases as their disposable income increases. (2)
- Aggregate demand increases. (1)
- The price level increases, but aggregate output remains unchanged in the long run. (2)



- Indicate in the diagram:
- AD shifts out (1)
- Vertical LRAS (1)
- P increases and Y unchanged (1)

- (a) Wealth effect: As many of the assets in the economy are denominated in nominal values, the higher the price level, the lower the purchasing power of money. This reduces the wealth of the economy (in real terms). As a result, households and firms reduce their purchases of all goods and services and the real output drops.

OR

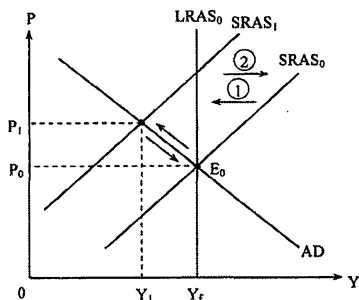
Interest rate effect: As the price level rises, households and firms demand more money to finance their transactions. Given the fixed supply of money, the interest rate would rise. An increase in interest rate would cause decline in investment and consumption and also the real output.

OR

Net exports effect: As the domestic price level rises, foreign-made goods become relatively cheaper so the quantity demanded of imports increases. However, the rise in the domestic price level also means that domestic-made goods are relatively more expensive to foreign buyers so the quantity demanded of exports decreases. When the volumes of exports decrease and of imports increase, net exports will decrease, so as the real output. (3)

- (b) Increase in the price of raw materials results in a rise in factor prices. So the SRAS will decrease. Price level increases and the output will fall in the short run.

The resulting excess supply of labour / deflationary (output) gap will result in a downward pressure on the input prices. In the long run, wages will adjust downward and the SRAS will increase and the price level will fall back to the original level. (5)



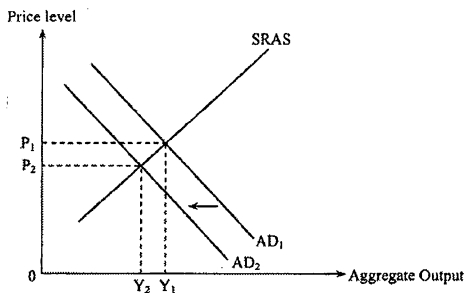
Indicate in the diagram:

- SRAS shifts to the left (1)
- P increases, Y drops (1)
- SRAS shifts back to the original position (1)
- final fall in P, back to the original level (1)

2013/DSE/II/12(c)

An upward adjustment of required reserve ratio reduces the money supply, resulting in an increase in interest rate and a drop in investment expenditure. (3)

The aggregate demand will fall, and so will the price level and output level. (2)



Indicate in the diagram:

- AD shifts to the left
- price level falls
- output level falls

(1)
(1)
(1)

2014/DSE/II/10(b)

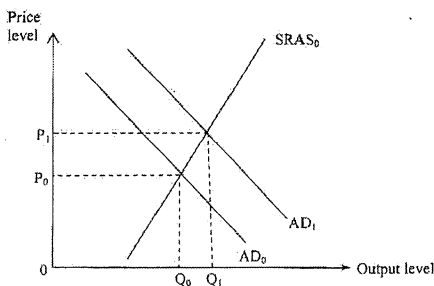
Relaxation of one-child policy would likely result in a rise in the birth rate, implying a bigger work force and a bigger aggregate supply and output in the future. (3)

2014/DSE/II/12(c)

The purchase of government bonds increases money supply, resulting in a drop in the interest rate. (2)

Investment / consumption expenditure increases and aggregate demand increases. (2)

In the short run, the output and price levels increases. (1)



Indicate on the diagram:

- AD curve shifts to the right
- higher price level
- higher output

(1)
(1)
(1)

2015/DSE/II/8

Proposal 1:

More money spent on education would increase government expenditure, leading to an increase in aggregate demand and thus aggregate output.

OR

Free education for longer years would improve labour productivity and thus aggregate output over time.

OR

Provision of free education beyond normal school age would reduce labour supply (by keeping more people at school) and thus aggregate output in the current period. (2)

Proposal 2:

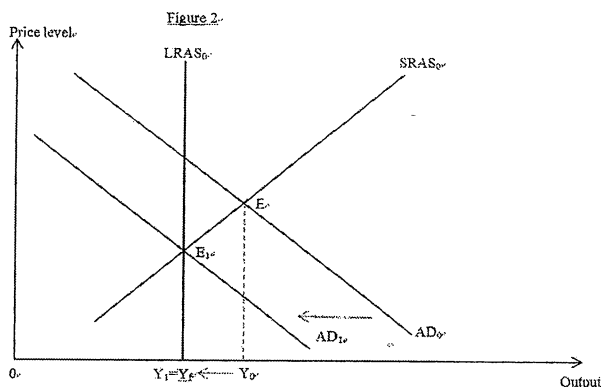
Unemployment benefits would increase the disposable income and consumption of the jobless, and thus aggregate demand, resulting in a rise in aggregate output.

OR

Unemployment benefits may, however, produce disincentive effects on labour supply, resulting in de-skilling of labour and a drop in aggregate output now and in the future. (2)

2015/DSE/II/12

- (b) Export would fall, (1)
 resulting in a drop in aggregate demand, (1)
 so output would fall back to Y_f . (1)



Indicate on the diagram:

- AD curve shifts to the left (1)
- correct position of initial short-run aggregate output Y_0 (1)
- correct position of new short-run aggregate output $Y_1 (=Y_f)$ (1)

- (c) Hong Kong is facing an inflationary gap. (1)
 There is an excess demand in the factor market, (1)
 creating pressure for factor prices to adjust upwards, (1)
 so that costs of production would increase over time and short-run aggregate supply (SRAS) would fall over time (, restoring output to Y_f in the long run). (1)

2016/DSE/II/5

When the price level increases, the adjustment of input prices is imperfect / incomplete (e.g. due to the long term contracts of factor of production), so the real cost of production -- i.e. nominal cost divided by the general price level -- will fall. Firms would thus use more factor inputs to produce larger output. Therefore, the SRAS curve is upward-sloping, i.e. a higher price level would result in a higher output level. (3)

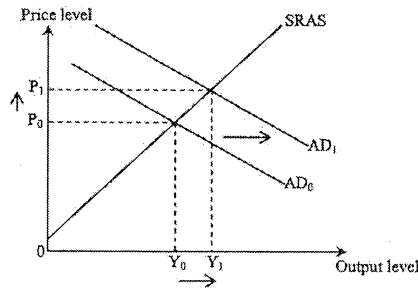
2016/DSE/II/12(b)

The cash-sharing scheme would raise the disposable income of the residents and therefore result in an increase in consumption.

(2)

Aggregate demand would rise, raising price and output levels accordingly.

(2)



Indicate in the diagram:

- rightward shift of AD

(1)

- higher P and Y

(1)

2017/DSE/II/9

(a) Depression / trough

(1)

(b) A reduction in interest rate would result in a rise in investment and/or consumption.

(1)

Aggregate demand would thus increase, and so would aggregate output as well.

(2)

(c) Consistent.

(1)

When interest rate dropped in 2009, GDP growth rate increased and unemployment rate dropped.

(3)

(d) - Subsidizing research and development

- importation of labour

- investment in infrastructures

(1)

[Mark the **FIRST** policy only.]

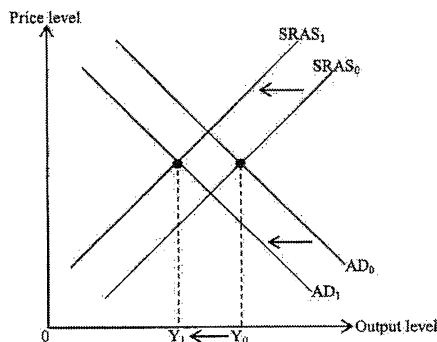
could increase potential output or production capacity of the economy as a whole in the long run, and thus shift LRAS rightwards.

(1)

2017/DSE/II/12(b)

The new visa requirement would increase labour cost for the British firm owners, resulting in a drop in SRAS.
 Political uncertainty would result in a drop in investment from the global investors and thus a fall in AD.
 Aggregate output would thus fall.

(2)
 (2)
 (1)



Indicate in the diagram:

- leftward shift of AD curve
- leftward shift of SRAS curve
- lower output

(1)
 (1)
 (1)

2017/DSE/II/13(b)

Proposal A's effect on aggregate output:

An increase in salaries tax would reduce AD via a drop in disposable (after-tax) income and thus consumption. As a result, aggregate output would fall.

[An increase in salaries tax may also reduce SRAS by creating a disincentive effect on labor.]

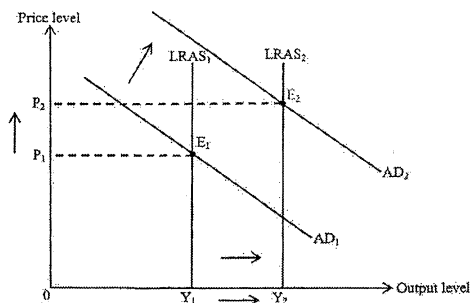
(4)

Proposal B's effect on aggregate output:

There would be no extra effect on aggregate output as a component of AD/AS does not change.

(2)

2018/DSE/II/9



Indicate in the diagram:

- rightward shift of AD (due to more visitors from the Mainland)
- rightward shift of LRAS (due to faster development of the HK economy)
- correct position of P_2 (at E_2), with a bigger horizontal shift in AD than in LRAS

(1)
 (1)

2018/DSE/II/11

The SRAS curve would decrease, as less labour is available for production.

(1)

The AD curve would also decrease, as the UN embargo reduces exports from Country Z.

(1)

As a result, aggregate output would fall.

(1)

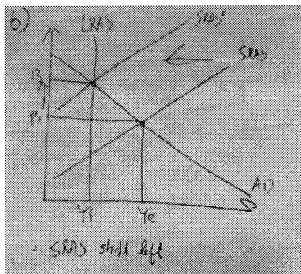
2018/DSE/II/13(a)(i)

Potential output would drop as the working-age population gradually decreases.

(2)

(from 5.36M in 2014 to 4.76M in 2054)

2020/DSE/II/6



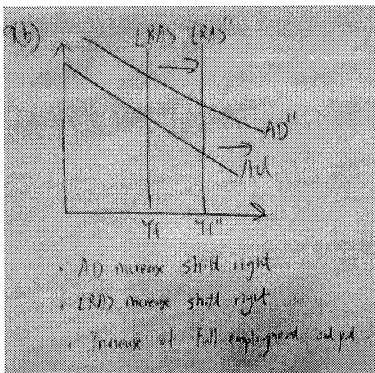
Inflationary gap exist as short run output is larger than full employment output.

Excess demand in labor market will drive up input price. (1 mark)

Production cost of firms increase and SRAS decrease and shift left (1 mark)

Short run output will decrease to full employment output level to close the inflationary gap (1 mark)

2020/DSE/II/9B



B) Infrastructure development will boost Government consumption expenditure (eg. Employment of more workers) and Gross investment expenditure (purchase of more capital good). Both G and I will increase the aggregate demand and the aggregate demand curve will shift rightward. (3 marks)

The construction of infrastructure will increase the quantity of factor endowment of an economy hence the production capacity will increase. LRAS increase and shift rightward. Full employment level will increase. (3 marks)

(a) Verbal elaborations:

Deflationary (output) gap refers to the difference between aggregate output (Y) and the full-employment output (Y_f) when Y is below Y_f . (1)

Illustrate in Figure 2:

– correct position of gap₀ (1)

(b) Verbal elaborations:

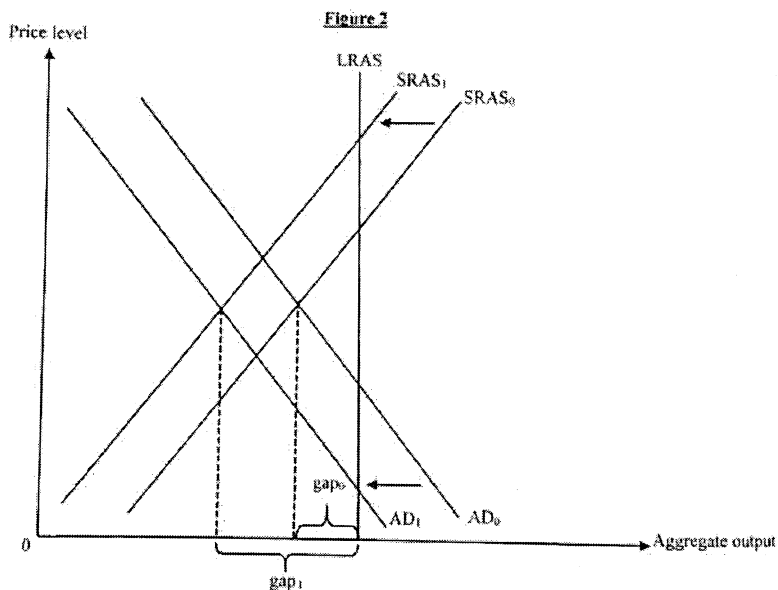
A temporary closure of factories would reduce aggregate supply in the short run. On the other hand, loss of jobs and pessimism would lower consumption (and possibly investment) expenditure and thus reduce aggregate demand. Together, shifts of the two curves would result in a drop in aggregate output (Y). Assuming, for simplicity, no change in long run aggregate supply and thus full-employment output (Y_f), then the fall in Y implies a wider deflationary gap. ⁶ (4)

Illustrate in Figure 2:

– leftward shift of SRAS curve (1)

– leftward shift of AD curve (1)

– correct position of gap₁ (1)

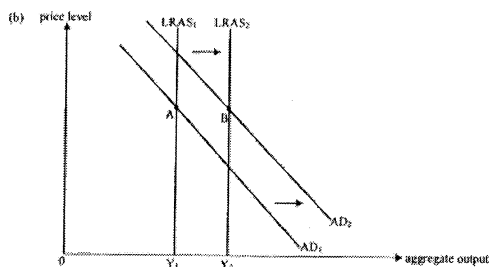


Marks

(4)

- (c) The central bank could use open-market purchase (i.e. buy government bonds from the public) to increase money supply. Then, the interest rate would fall, thus inducing firms to invest more, implying an increase in aggregate demand. As a result, aggregate output (Y) would rise in the short run, but long-run output (Y_L) would remain unchanged, so that the deflationary gap ($= Y - Y_L$) would become narrower.

2022/DSE/II/9b



Illustrate in the diagram:

- arightward shift in the AD curve (1)
- arightward shift in the LRAS curve (1)
- ahigher level of aggregate output (1)

Verbal elaboration:

Better infrastructure would lead to an increase in production capacity and potential Output and thus an increase in long-run aggregate supply. More infrastructure investment would lead to an increase in aggregate demand. Aggregate output would hence increase. (5)

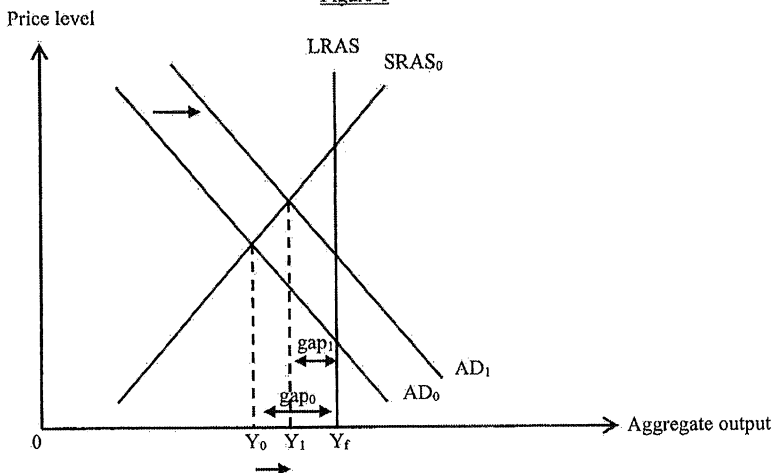
2023/DSE/II/7

(a) Indicate in the diagram:

- correct position of vertical LRAS curve and deflationary (output) gap

(1)

Figure 1



(b) Verbal elaboration:

- As more travellers visit the country, export of services increases.
- Aggregate demand increases.
- Aggregate output increases, narrowing the deflationary (output) gap.

(1)

(1)

(1)

Indicate in the diagram:

- rightward shift of AD curve
- increase in Y , decrease in deflationary (output) gap

(1)

(1)

2023/DSE/II/12c

- (c) Increase in discount rate raises the cost for commercial banks to borrow money from the Federal Reserve. Monetary base tends to decrease. (1)
- Money supply decreases. (1)
- Interest rates increase.¹ (1)
- Investment decreases. (1)
- Aggregate demand decreases. Price level tends to decrease, relieving the pressure on the rising price level. (1)

(a) Reasons:

- Wealth effect: As many of the assets in the economy are denominated in nominal values, the higher the price level, the lower the purchasing power of these monetary assets. This reduces the wealth of the economy (in real terms). As a result, consumption decreases and the aggregate quantity of output demanded drops.
- Interest rate effect: As the price level rises, households and firms demand more money to finance their transactions. Given the fixed supply of money, the interest rate would rise. An increase in interest rate would cause a decline in investment and consumption and thus the aggregate quantity of output demanded.
- Net exports effect: As the domestic price level rises, foreign-made goods become relatively cheaper so the quantity demanded of imports increases. The rise in the domestic price level also means that domestically produced goods are relatively more expensive to foreign buyers, so the quantity demanded of exports decreases. When the volume of exports decreases and that of imports increases, net exports will decrease (if the price elasticities of demand for exports and/or imports are sufficiently high). As net exports are a component of aggregate expenditure, the aggregate quantity of output demanded would drop as net exports decline.

3@
max: 3

[Mark the FIRST point ONLY.]

(b) Verbal elaboration:

The economy is facing a deflationary (output) gap. (1)

There is an excess supply in the factor market. (1)

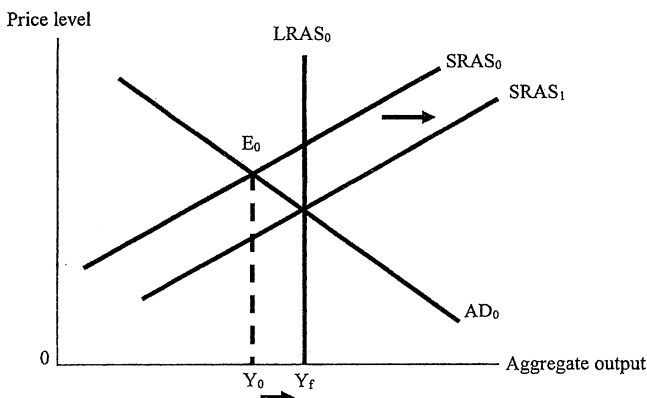
It creates pressure for factor prices to adjust downward. Costs of production would drop. (1)

Short run aggregate supply (SRAS) would increase over time, (1)

thus restoring output level to the full-employment output level, Y_f , in the long run.

Indicate in the diagram:

- Rightward shift of short run aggregate supply curve, reaching Y_f . (1)



2024/DSE/II/12(a)

(a) Verbal elaboration:

- Due to the construction, gross investment expenditure will increase.
- Aggregate demand will increase.
- Aggregate output will increase in the short run.

(1)
(1)
(1)

Indicate in the diagram:

- rightward shift of aggregate demand curve
- increase in aggregate output

(1)
(1)

